

WELL COMPLETION REPORT (SYNTHETIC)

OIL-MRN-009 - Moran-009

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000099
Field / District	Moran / Dibrugarh
Block	PEL-ASSAM-03
Basin	Assam-Arakan
Formation	Lakadong+Therria (LKT)
Spud / Completion	2005-02-08 / 2005-06-13
Measured / TVD	3770 m / 3737 m
Trajectory	Deviated; inc 8.0 deg; az 310.6 deg
Formation Interval	2190-3040 m
Lithology	Sandstone with carbonaceous interbeds
Pore Pressure Reference	3914 psi
Fracture Gradient	14.43 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Critical (64.0/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	942	Surface
9 5/8 in casing	2186	Intermediate
7 in liner/casing	3317	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2005-03-30 1668 m	Stuck Pipe	Critical	High drag was recorded while circulating remained possible. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2005-04-20 2264 m	Cementing Issue	Critical	Cement returns were below planned percentage. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Volume imbalance is an important cement-placement warning.
2005-03-03 860 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2005-03-07 979 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Volume imbalance is an important cement-placement warning.
2005-03-10 1082 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2005-04-01 1724 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2005-04-01 1738 m	Kick	High	Drilling break followed by positive flow check. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.

Completion Summary

The synthetic well reached 3770 m MD in the Lakadong+Therria interval. The NWIS knowledge record links 15 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.